CANDIDATE – PLEASE NOTE!

PRINT your name on the line below and return this booklet with your answer sheet. Failure to do so may result in disqualification.

FORM TP 2019018

TEST CODE **01234010**

JANUARY 2019

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN SECONDARY EDUCATION CERTIFICATE® EXAMINATION

MATHEMATICS

Paper 01 – General Proficiency

1 hour 30 minutes

04 JANUARY 2019 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This test consists of 60 items. You will have 1 hour and 30 minutes to answer them.
2. In addition to this test booklet, you should have an answer sheet.
3. **A list of formulae is provided on page 2 of this booklet.**
4. Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer, and decide which choice is best.
5. On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

   Sample Item

   $2a + 6a =$

   (A) $8a$
   (B) $8a^2$
   (C) $12a$
   (D) $12a^2$

   Sample Answer

   ● (B) (C) (D)

   The best answer to this item is "$8a$", so (A) has been shaded.

6. If you want to change your answer, erase it completely before you fill in your new choice.
7. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.
8. You may do any rough work in this booklet.
9. Figures are not necessarily drawn to scale.
10. Calculators and mathematical tables are NOT allowed for this paper.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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1. $\left(\frac{2}{3}\right)^2$ is equal to

(A) $\frac{4}{6}$

(B) $\frac{4}{3}$

(C) $\frac{2}{9}$

(D) $\frac{4}{9}$

2. The number 3.14063 written correct to 3 decimal places is

(A) 3.140
(B) 3.141
(C) 3.146
(D) 3.150

3. In standard notation, 208.06 is written as

(A) 2.0806×10^2
(B) 20.806×10^1
(C) 2.0806×10^{-2}
(D) 0.20806×10^3

4. If \$560 is shared in the ratio 2:3:9, the difference between the largest and the smallest shares is

(A) \$ 80
(B) \$240
(C) \$280
(D) \$360

5. If 60% of a number is 90, what is $\frac{4}{5}$ of the number?

(A) 54
(B) 72
(C) 120
(D) 150

6. The first three common multiples of 3, 4 and 6 are

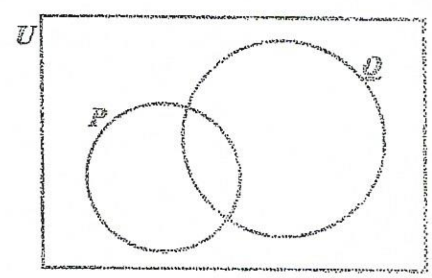
(A) 0, 1, 2
(B) 3, 4, 6
(C) 6, 8, 12
(D) 12, 24, 36

7. If $P = \{2, 3, 5, 7\}$,
$Q = \{2, 3, 6\}$ and
$S = \{2, 4, 5\}$

then $P \cap Q \cap S =$

(A) $\{2\}$
(B) $\{2, 3\}$
(C) $\{2, 3, 5\}$
(D) $\{2, 3, 4, 5, 6, 7\}$

<u>Item 8</u> refers to the following Venn diagram.



In the Venn diagram, $n(P) = 5$, $n(Q) = 9$ and $n(P \cup Q) = 10$.

8. What is $n(Q \cap P)$?

(A) 1
(B) 4
(C) 5
(D) 9

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9. Which of the following sets is equivalent to the set $\{p, q, r, s\}$?

(A) $\{4\}$
(B) $\{p, q, r\}$
(C) $\{a, b, c, d\}$
(D) $\{1, 2, 3, 4, 5\}$

10. All students in a class play Scrabble or Checkers or both. If 36% of the students play Scrabble only and 48% of the students play Checkers only, what percentage of students play BOTH games?

(A) 12
(B) 16
(C) 84
(D) 88

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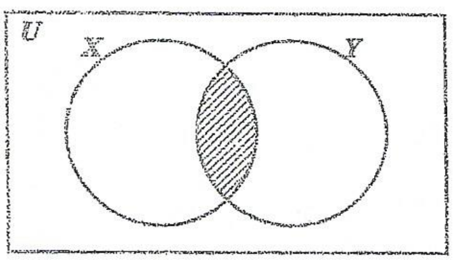
Item 11 refers to the following Venn diagram.



11. In the Venn diagram, X represents the set of factors of 12. Y represents the set of factors of 9. The shaded region represents the set of all factors of

(A) 3
(B) 6
(C) 21
(D) 108

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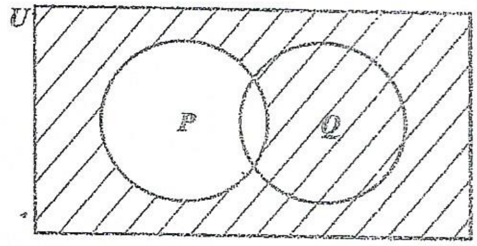
Item 12 refers to the following Venn diagram.



12. In the Venn diagram, the shaded area represents

(A) P'
(B) $(P \cup Q)'$
(C) $Q \cup P'$
(D) $Q \cap P'$

13. A store offers a discount of 15% on the total bill to customers who spend more than \$20. If a customer's total bill is \$80, how much must she pay?

(A) \$60
(B) \$65
(C) \$68
(D) \$71

14. At a store, sales tax is charged at a rate of 2% on the cost price of an item. The sales tax on a dress which costs \$180 is

(A) \$ 3.60
(B) \$ 10.80
(C) \$176.40
(D) \$183.60

15. A plot of land is valued at $18 000. Land tax of $126 is charged annually. At what rate per annum is the land tax charged?

(A) 0.07%
(B) 0.7%
(C) 7%
(D) 70%

16. The exchange rate for one United States dollar (US$1.00) is two dollars and seventy cents in Eastern Caribbean currency (EC$2.70). What is the value of US$4.50 in EC currency?

(A) $ 1.67
(B) $ 6.00
(C) $ 7.20
(D) $12.15

17. A man's annual salary is $45 000. His tax free allowances total $13 000. He has to pay a tax of 20% on his taxable income.

The tax payable is

(A) $ 2 600
(B) $ 6 400
(C) $ 9 000
(D) $11 600

18. A television set costs $350 cash. When bought on hire purchase, a deposit of $35 is required, followed by 12 monthly payments of $30. How much is saved by paying cash?

(A) $10
(B) $25
(C) $40
(D) $45

19. A man pays 60 cents for every 200 m^3 of gas used, plus a fixed charge of $13.75. How much does he pay when he uses 55 000 m^3 of gas?

(A) $178.75
(B) $175.25
(C) $165.00
(D) $151.25

Item 20 refers to the following table which shows the rates charged by an insurance company.

House Insurance	25¢ per $100
Contents Insurance	50¢ per $100

20. How much will a person pay for insurance, if his house is valued at $50 000, and the contents at $10 000 ?

(A) $175
(B) $225
(C) $450
(D) $500

21. The product of $(-2a)$ and $(-3a)$ is

(A) $-6a$
(B) $-5a$
(C) $5a^2$
(D) $6a^2$

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GO ON TO THE NEXT PAGE

22. When 6 is added to a number and the sum is divided by 3, the result is 4. This statement written using mathematical symbols is

(A) $\frac{6}{3} + x = 4$

(B) $\frac{6+x}{3} = 4$

(C) $\frac{6+x}{3} = \frac{4}{3}$

(D) $6 + \frac{x}{3} = 4$

23. Althea normally saves \$$x$ each month; but in June she saved \$4 more than twice her usual amount. In June she saved

(A) \$ $4x$
(B) \$ $6x$
(C) \$ $(x + 4)$
(D) \$ $(2x + 4)$

24. If $3 + \frac{2}{x} = 1$, the value of x is

(A) -1

(B) $\frac{1}{5}$

(C) $\frac{1}{2}$

(D) 5

25. If $x = 4$ and $y = 2$, what is the value of

$\frac{x^2 + 3y}{xy}$?

(A) $1\frac{3}{4}$

(B) $2\frac{1}{2}$

(C) $2\frac{3}{8}$

(D) $2\frac{3}{4}$

26. The expression $-2(x - 4)$ is the same as

(A) $-2x - 8$
(B) $-2x + 8$
(C) $-2x - 4$
(D) $-2x + 4$

27. If $|A| = 0$, then A is

(A) an inverse matrix
(B) a singular matrix
(C) an identity matrix
(D) a non-singular matrix

Item 28 refers to the following matrices, X and Y.

$$X = \begin{pmatrix} 1 & 3 & -3 \\ 3 & 0 & 5 \end{pmatrix}, \quad Y = \begin{pmatrix} 3 & 0 \\ 2 & 1 \\ 0 & 5 \end{pmatrix}$$

28. What is the order of the matrix product XY?

(A) 2×2
(B) 2×3
(C) 3×2
(D) 3×3

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29. If the vectors **p** and **q** are $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ respectively, then $-\mathbf{p} - 2\mathbf{q}$ is

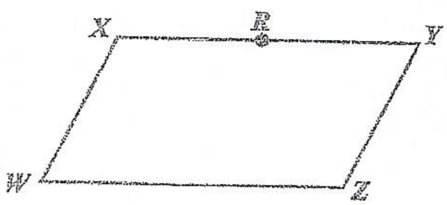
(A) $\begin{pmatrix} 5 \\ -6 \end{pmatrix}$

(B) $\begin{pmatrix} -2 \\ -6 \end{pmatrix}$

(C) $\begin{pmatrix} 5 \\ 10 \end{pmatrix}$

(D) $\begin{pmatrix} -1 \\ -10 \end{pmatrix}$

Item 30 refers to the following parallelogram, *WXYZ*. In the parallelogram, *R* is the midpoint of *XY*.



30. Based on the information above, $\overrightarrow{ZR}$ =

(A) $\overrightarrow{WX} + \frac{1}{2}\overrightarrow{WZ}$

(B) $\overrightarrow{WX} - \frac{1}{2}\overrightarrow{WZ}$

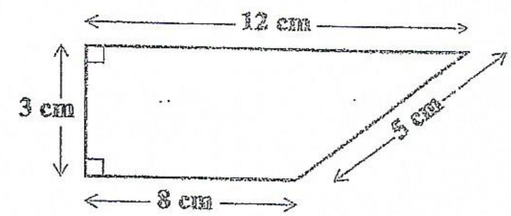
(C) $\frac{1}{2}\overrightarrow{WX} - \overrightarrow{WZ}$

(D) $\frac{1}{2}\overrightarrow{WX} + \overrightarrow{WZ}$

31. If it took a speedboat 9 hours to travel a distance of 1 080 km, what was its average speed, in kmh^{-1}?

(A) 12
(B) 102
(C) 120
(D) 1200

Item 32 refers to the following diagram of a trapezium.



32. The area of the trapezium is

(A) 24 cm²
(B) 28 cm²
(C) 30 cm²
(D) 36 cm²

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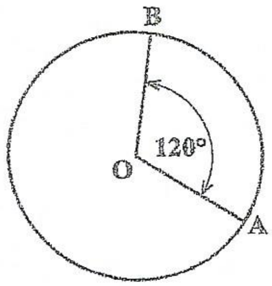
Item 33 refers to the following circle, with centre O.



33. If the circumference of the circle is 15 cm, then the length of the minor arc AB, in cm, is

(A) $\frac{360}{120} \times 15$

(B) $\frac{120}{360} \times 15$

(C) $\frac{360}{360 - 120} \times 15$

(D) $\frac{360 - 120}{360} \times 15$

34. The lengths of the sides of a triangle are x, $2x$ and $2x$ centimetres. If the perimeter is 20 centimetres, what is the value of x?

(A) 4
(B) 5
(C) 8
(D) 10

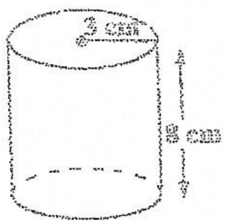
35. A rectangular garden plot, 15 m long and 12 m wide, is used to plant fruits and vegetables. If 80 m^2 of the plot is used to plant vegetables, what area of the plot is planted with fruits?

(A) 26 m^2
(B) 100 m^2
(C) 134 m^2
(D) 260 m^2

36. The distance around a lake is 8 km. On a map, this distance around the lake is represented by a length of 2 cm. The scale on the map is

(A) 1:40
(B) 1:2 000
(C) 1:200 000
(D) 1:400 000

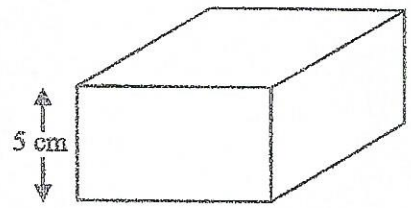
Item 37 refers to the following diagram.



37. The diagram above, not drawn to scale, shows a cylinder of radius 3 cm and height 8 cm. The volume of the cylinder is

(A) 12π cm^3
(B) 48π cm^3
(C) 72π cm^3
(D) 192π cm^3

Item 38 refers to the following diagram, **not drawn to scale**, which shows a cuboid.



38. The volume of the cuboid is 320 cm^3 and the height is 5 cm. If the cuboid has a square base, what is the length of one side of the base?

(A) 8 cm
(B) 16 cm
(C) 32 cm
(D) 64 cm

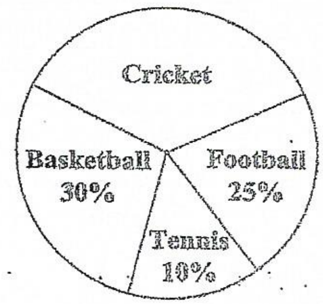
Item 39 refers to the following scores which were obtained by 11 students in a competition.

5, 3, 6, 8, 7, 8, 3, 11, 6, 3, 3

39. The modal score is

(A) 3
(B) 6
(C) 8
(D) 11

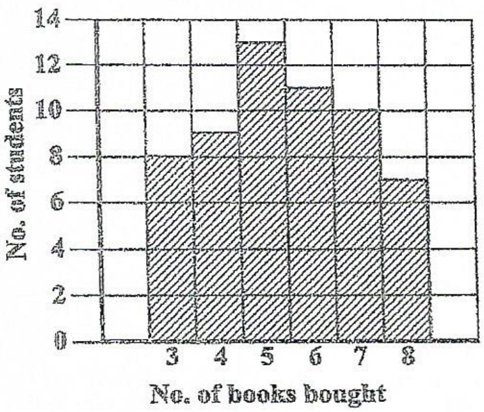
Item 40 refers to the following pie chart which shows the popular games played by 720 students.



40. How many students play cricket?

(A) 35
(B) 120
(C) 252
(D) 300

<u>Items 41–42</u> refer to the following chart which shows the number of books 58 students bought at a sale.



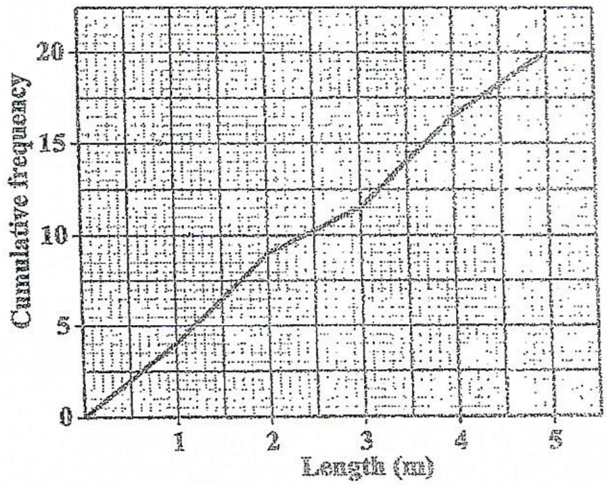
41. How many students bought exactly 4 books?

(A) 8
(B) 9
(C) 10
(D) 14

42. The median number of books bought by the students is

(A) 5
(B) 6
(C) 29
(D) 30

<u>Items 43–44</u> refer to the following diagram, which shows the cumulative frequency polygon of the lengths, in metres, of 20 fish that were caught by two fishermen.



43. The semi-interquartile range of the lengths of the fish is

(A) 1.20 m
(B) 1.25 m
(C) 2.5 m
(D) 3.7 m

44. A fish is selected at random. What is the probability that it is at LEAST 1.8 m long?

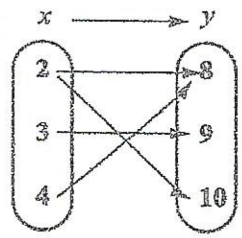
(A) $\frac{5}{20}$

(B) $\frac{8}{20}$

(C) $\frac{12}{20}$

(D) $\frac{15}{20}$

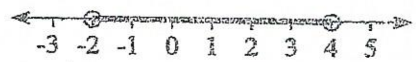
Item 45 refers to the following arrow diagram.



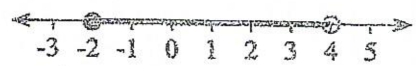
45. The arrow diagram above describes the relation

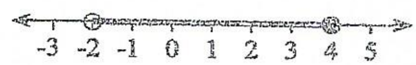
(A) y is less than x
(B) y is a factor of x
(C) x is a factor of y
(D) x is a multiple of y

46. Which of the following line graphs represents $\{ x : -2 < x \leq 4 \}$?

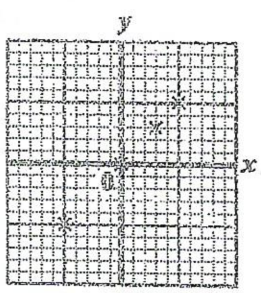
(A)


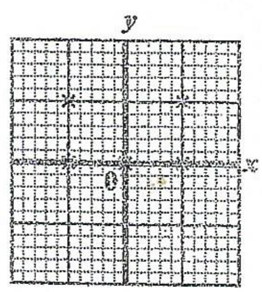
(B)

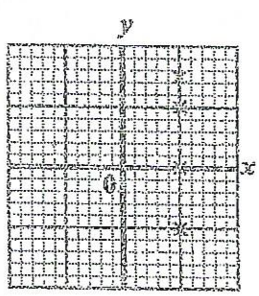

(C)


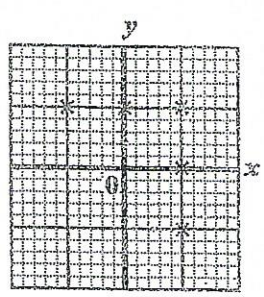
(D)


47. Which of the following graphs represents a function?

(A)


(B)


(C)


(D)


48. Which of the following points lies on the line $y = 2x - 3$?

(A) (2, 3)
(B) (–2, –1)
(C) (4, 1)
(D) (0,–3)

49. If $f(x) = 2x^2 - 1$, then $f(-3) =$.

(A) –32
(B) –19
(C) 17
(D) 35

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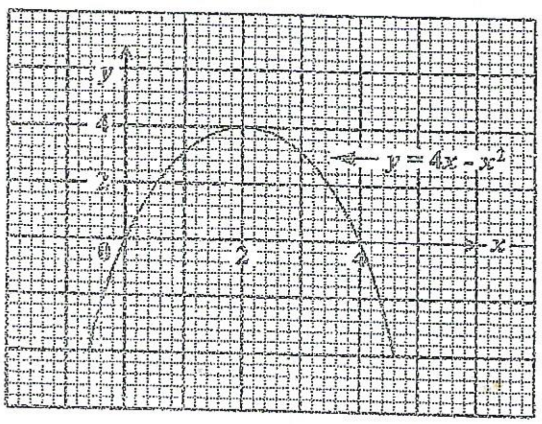
Items 50–51 refer to the following graph.



50. The maximum point of $y = 4x - x^2$ is

(A) (0, 0)
(B) (0, 4)
(C) (2, 4)
(D) (4, 2)

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51. The values of x at the points where $y = 4x - x^2$ intersects $y = 0$ are

(A) $x = 0$ and $x = 4$
(B) $x = 0$ and $x = 2$
(C) $x = 0$ and $x = -4$
(D) $x = 2$ and $x = 4$

52. Which of the following sets is represe by the function f: $x \rightarrow x^2 + 3$ w $x \in \{0, 1, 2, 3\}$?

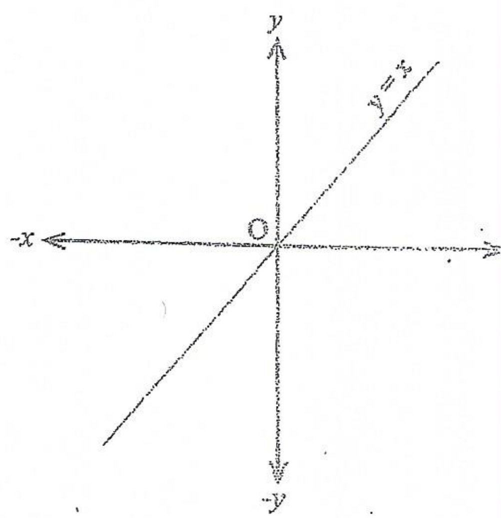
(A) {(0, 3), (1, 4), (2, 7), (3, 12)}

(B) {(0, 3), (1, 5), (2, 7), (3, 9)}

(C) {(0, 3), (1, 4), (2, 5), (3, 6)}

(D) {(0, 3), (1, 1), (2, 4), (3, 9)}

Item 53 refers to the following diagra

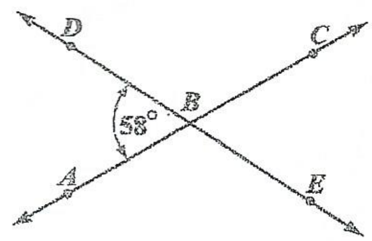


53. In the diagram above, if the line $y =$ rotated anti-clockwise about O through what is its image?

(A) $y = 0$
(B) $x = 0$
(C) $y = x$
(D) $y = -x$

Items 54–55 refer to the following diagram of a pair of intersecting lines.

AC and DE are straight lines intersecting at B. Angle $DBA = 58°$.



54. The measure of angle ABE is

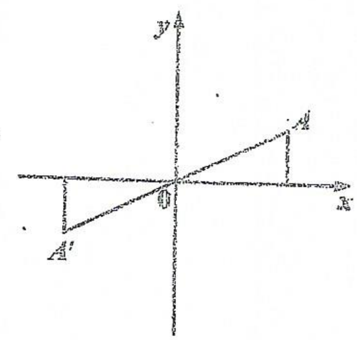
(A) 58°
(B) 122°
(C) 142°
(D) 302°

55. Which of the following angles are equal?

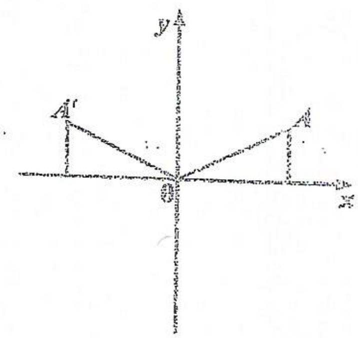
(A) $\angle DBC$ and $\angle CBE$
(B) $\angle CBE$ and $\angle ABE$
(C) $\angle ABD$ and $\angle CBD$
(D) $\angle ABD$ and $\angle CBE$

56. In each of the following diagrams, A' is image of A. Which of the diagrams sho a reflection in the x–axis?

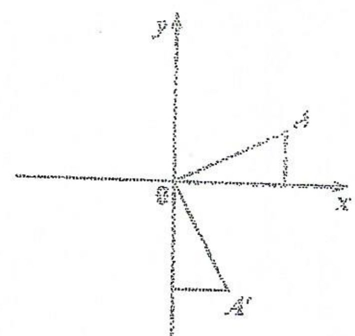
(A)



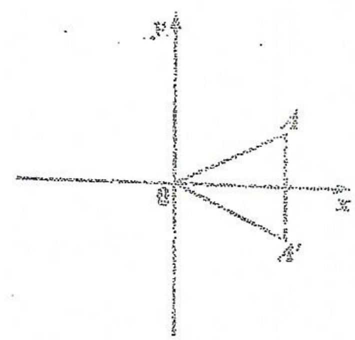
(B)



(C)



(D)



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57. In triangle ABC, angle $A = x°$ and angle $B = 2x°$. What is the size of angle C?

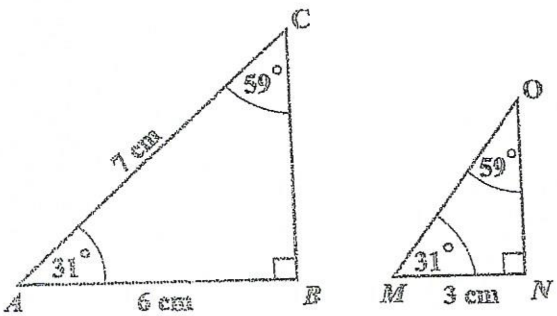
(A) $(180 - 3x)°$

(B) $60°$

(C) $30°$

(D) $\left(\frac{180}{3x}\right)°$

Item 58 refers to the following pair of similar triangles.

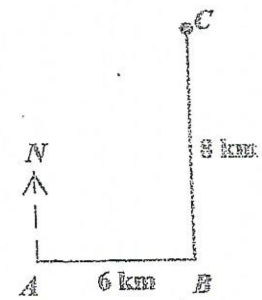


58. The length of MO, in centimetres, is

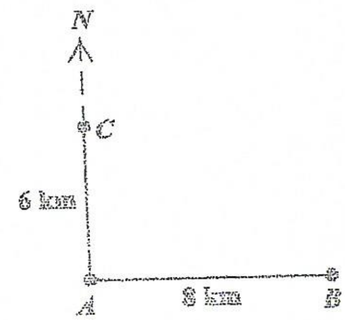
(A) 3.0
(B) 3.5
(C) 6.0
(D) 10.5

59. A ship sailed 8 km due east from A to B. It then sailed 6 km due north to C. Which of the following diagrams BEST represents the path of the ship?

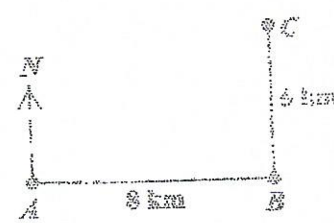
(A)



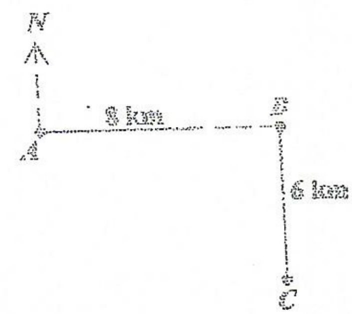
(B)



(C)



(D)



60. Under the translation $\begin{bmatrix} 2 \\ -3 \end{bmatrix}$, the image of (3, –5) is

(A) (0, –3)
(B) (1, –2)
(C) (5, –8)
(D) (6, 15)

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A003

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.